

Matgeo-5.13.12

AI25BTECH11036-SNEHAMRUDULA

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The set of all values of λ for which the system of linear equations

$$2x_1 - 2x_2 + x_3 = \lambda x_1,$$

$$2x_1 - 3x_2 + 2x_3 = \lambda x_2,$$

$$-x_1 + 2x_2 = \lambda x_3,$$

has a non-trivial solution

- (a) contains two elements
- (b) contains more than two elements
- (c) is an empty set
- (d) is a singleton

$$A = \begin{pmatrix} 2 & -2 & 1 \\ 2 & -3 & 2 \\ -1 & 2 & 0 \end{pmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & -2 & 1 \\ 2 & -3 - \lambda & 2 \\ -1 & 2 & -\lambda \end{vmatrix}$$

$$= -\lambda^3 - \lambda^2 + 5\lambda - 3$$

$$\lambda^3 + \lambda^2 - 5\lambda + 3 = 0$$

$$(\lambda - 1)^2(\lambda + 3) = 0$$

$$\lambda_1 = -3, \quad \lambda_2 = 1, \quad \lambda_3 = 1$$

$$(A + 3I)v = 0 \quad \Rightarrow \quad \begin{pmatrix} 5 & -2 & 1 \\ 2 & 0 & 2 \\ -1 & 2 & 3 \end{pmatrix} v = 0$$

$$v_1 = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$$

$$(A - I)v = 0 \quad \Rightarrow \quad \begin{pmatrix} 1 & -2 & 1 \\ 2 & -4 & 2 \\ -1 & 2 & -1 \end{pmatrix} v = 0$$

solution III

$$v_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$P = \begin{pmatrix} -1 & 2 & -1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} -3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P^{-1} = \begin{pmatrix} \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{pmatrix}$$

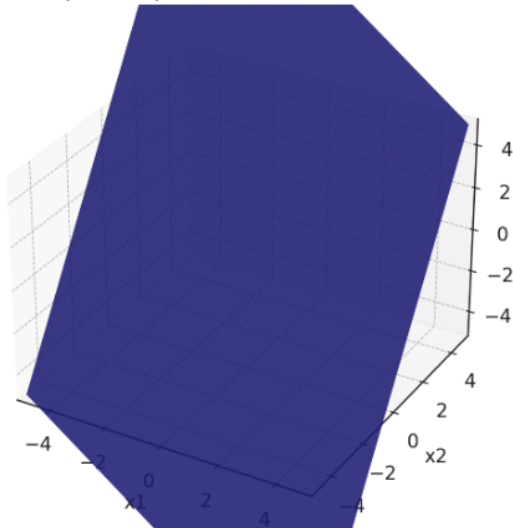
$$A = PDP^{-1}$$

$$\{\lambda : \det(A - \lambda I) = 0\} = \{-3, 1\}$$

(A) Contains two elements.

Graphical Representation

3D plot of planes for $\lambda \approx 1.000000$



Graphical Representation

3D plot of planes for $\lambda \approx -3.000000$

